

Advait Patel

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EDUCATION

University of Illinois - Urbana-Champaign

Champaign, IL

BS in Computer Science – GPA: 3.8/4.0

Aug. 2022 – December 2025

Relevant Coursework: Real-time Systems, Machine Learning, Deep Learning for Robotics (Grad), Algorithms, Probability/Stats, Data Structures, Systems Programming, Computer Vision (Grad), Linear Algebra

SKILLS

Rust, Python, PyTorch, C, C++, Amazon Web Services (AWS), Kotlin, OpenCV, SQL, JavaScript, Typescript, Git/GitHub, Flask, React, PyTorch, Java, Keras, Pandas, NumPy, Matplotlib, Docker, Linux/Ubuntu, ROS

EXPERIENCE

Anduril Industries

May 2025 – August 2025

Software Engineer Intern

Costa Mesa, CA

- Implemented bidirectional streaming in Rust for RAIL, Anduril's communication/hardware abstraction layer.
- Developed performance benchmarks and optimizations in Rust for latency and throughput in Rust.
- Integrated benchmarking suite with automated CI tests and with Palantir Foundry for visualization.

University of Illinois

November 2024 – Present

Undergraduate Researcher

Champaign, IL

- Exploring evaluation of robotics policies through Real2Sim using 3D Gaussian Splatting and MuJoCo.
- Aligned point clouds between real world, Gaussian Splat, and MuJoCo to create a realistic simulation environment.
- Developed end-to-end pipeline for evaluating robot learning policies using a custom hybrid rendering method.

Capital One

June 2024 - August 2024

Software Engineer Intern

McLean, VA

- Created a divisional-level dashboard to monitor problems with teams' AWS usage using Typescript, React, and AWS.
- Decreased loading time on application page by 10x by optimizing GraphQL queries.
- Implemented AWS Lambda with Python and SQLAlchemy to load data from 5 different sources into PostgreSQL.

Yield Protocol

May 2023 – August 2023

Software Engineer Intern

Remote

- Developed a Web3 chatbot using open-source large language models and Python LangChain API integrations, resulting in 20% cost reduction compared to OpenAI APIs.
- Finetuned SOTA large language models on enterprise-grade hardware with AWS Sagemaker and Huggingface APIs.
- Created a custom question-answer pipeline dataset for model finetuning, resulting in over 4,000 unique data points, which enhanced model finetuning and increased response accuracy by 35%.

LEADERSHIP/EXTRACURRICULARS

SIGRobotics (ACM @ UIUC) | Founder, Chair | Python, ROS 2, PyTorch, MuJoCo

April 2024 – Present

- Trained pick-and-place imitation learning policies using PyTorch, LeRobot, and the Alex Koch arms.
- Developed and open-sourced LeKiwi, a low-cost mobile manipulator in collaboration with Huggingface/LeRobot.

UIUC Gies Disruption Lab | Technical Lead | Python, Huggingface, AWS

January 2023 – December 2023

- Led a team of 6 software engineers to develop an AI tutor specific to UIUC courses using finetuned open-source LLMs, vector databases, and LangChain API integrations.

PROJECTS

AI Chess Pipeline | OpenCV, Python, PyTorch, OpenAI

- Developed a pipeline that takes in a top down view of a chess board and determined board state using a Resnet-18.
- Developed an AI-assisted data annotation platform using GPT-4o-mini in order to label 1.5k images.

Robot Hackathon Project | Python, PyTorch, Huggingface

- Won the second place prize and the Fashion Award at the NYC Robot Arm Hackathon.
- Built a T-shirt folding station by training an ACT imitation learning policy on 85 demonstrations using LeRobot.
- Modified LeRobot library from Huggingface to allow a bimanual arm setup.